PRESENCIFY: A CPU Only Face Recognition Attendance Prototype with Fixed Confidence Thresholds and an Uncertainty Review Band

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Abstract: *PRESENCIFY proposes a novel edge-oriented, privacy-preserving face recognition attendance prototype designed to run on standard laptop CPUs without any dedicated graphics accelerator. This pipeline uses MediaPipe BlazeFace for fast face detection in tandem with a lightweight EdgeFace XS feature extractor, yielding 512-dimensional L2-normalized embeddings compared using cosine similarity. It uses a SQLite storage mechanism and Streamlit administrative interface. This study examines how fixed decision thresholds and uncertainty review bands manage classification mistakes on one hand and human workloads on another hand. Empirical evaluation on 166 static probe images demonstrates that decision threshold of 0.65 successfully authenticates 65 true positives but mistakenly classifies 7 true positives. However, the use of a single decision threshold allows 18 impostors to pass the authentication, creating a 27.27% false acceptance rate. Staged decision process channels 56 uncertain cases to uncertainty review band which successfully catches 33 impostors at the expense of sending 33.73% cases for human verification. Simulated oracle review process indicates that ideal human decisions could restore 12 genuine student records, bringing total accuracy up to 75.30%. Significantly, this architecture does not have any presentation attack detection (PAD_UNAVAILABLE). Image captures contain benchmark replacement of initial subjects. The prototype thus presents baseline evidence on the edge of automated throughput and administrative workload.*

Index Terms: face recognition, attendance system, edge computing, confidence thresholds, human review, presentation attack detection.

I. INTRODUCTION

Traditional attendance roll-calls during university lectures are time-consuming and highly susceptible to cheating through proxy attendance by fellow students. While computer vision methods offer a promising solution for automating classroom verification [1]–[3], real-world deployment of attendance systems in academia faces hardware limitations. Lectures usually occur in department halls that use modest dual or quad core laptops with no graphics processing unit. Thus, campus attendance kiosks should include compact biometric architectures capable of running on commodity hardware without transmitting sensitive student facial information to any distant servers.

Biometric feature extractors project facial crops in a metric embedding space, producing similarity scores through cosine similarity but not true posterior probabilities of personal identification. In such a case, usage of a single static decision threshold is inevitably associated with certain operational trade-offs. On the one hand, a strict decision threshold guarantees high security from unauthorized access but leads to numerous false rejections of legitimate students; on the other hand, a loose decision threshold decreases classroom disruptions but allows unauthorized people to sign in unnoticed. Diversion of uncertain scores to uncertainty review band may serve as an operational buffer, but this triaging strategy immediately shifts inspection responsibilities to the teaching personnel. An important question is whether fixed confidence thresholds and human review queues can guarantee security without creating an excessive burden for teachers.

This paper attempts to answer this question with the help of evidence-based realization and audit of PRESENCIFY. The project makes three main contributions. First, it proposes a complete offline attendance workflow optimized for commodity x86 CPU architectures without any GPU dependencies. Second, it systematically estimates false accepts, false rejects, and human inspection workloads depending on similarity intervals. Third, it distinctly separates automated system performance from counterfactual improvements due to idealized oracle review.

II. LITERATURE REVIEW

A. Lightweight recognition and detection

State-of-the-art edge biometric architectures focus on development of model compression approaches for fast facial feature extraction on edge devices [4]–[6]. George et al. presented EdgeFace, which uses hybrid convolutional transformer backbone to compress deep face recognition models to low-memory checkpoints for edge hardware [4]. PRESENCIFY utilizes an EdgeFace XS checkpoint (gamma = 0.6) to produce compact 512-dimensional representations. Setyawan et al. created FaceLiViT which uses linear vision transformers and reparameterization technique to speed up mobile inference [5]. Surveys stress that edge benchmarks need to provide real inference latencies in addition to standard verification accuracies [6]. Taxonomical classification of biometric architectures is done based on feature discriminability, operational modality, and embedding dimensionality [7]. Despite advanced architectures such as YOLO-FaceV2 being capable of robust scale-aware detection with severe occlusion [8], PRESENCIFY relies on MediaPipe BlazeFace due to lighter CPU execution footprint.

B. Facial recognition attendance systems

Automatic classroom attendance is the subject of intensive academic research [1]–[3]. Ahmad et al. compared several pretrained deep neural networks embedded into an attendance GUI [1]. Patel et al. investigated multi-student surveillance tracking via fixed CCTV cameras coupled with deep convolutional models [2]. Likewise, Fan coupled MobileNet-based feature extraction with sparrow search support vector

machine to allocate resources optimally in the classroom [3]. Recent bibliometric analysis shows increased attention to biometrics in education institutions [9]. Since previously reported systems utilize heterogeneous sensor inputs, varying classroom lighting conditions, and proprietary test datasets, direct quantitative comparison of numerical scores among various works is empirically flawed. PRESENCIFY concentrates on transparency in threshold selection, auditing of the review queue, and honest reporting of errors.

Attendance kiosks work in extended class sessions and not with isolated frames. Robust deployments must reduce duplications of the same detection across adjacent video frames, keep immutable session logs, and not allow unvalidated detection to overwrite validated records. Lighting changes, camera perspective angles, and students' facial expressions cause perturbations in cosine similarity scores. Evaluation of two-threshold acceptance policy thus requires assessment of borderline case audits, instructor review workload, and sustained high-confidence false accepts.

Successive video frames obtained in fractions of seconds demonstrate huge temporal correlation and cannot be treated as statistically independent trials. PRESENCIFY solves this problem at run time by anchoring attendance to session windows and student IDs. While such implementation streamlines record keeping, it cannot overcome degraded feature representation or act as a replacement for physical spoof detection.

C. Presentation attacks, privacy and fairness

Face verification should be distinguished from active presentation attack detection (PAD) and anti-spoofing protection measures [10]–[13]. Exhaustive survey shows that protecting against presentation attacks requires special optical sensors, customized neural network heads, and testing against numerous PAD techniques [10]–[12]. Recent methodology investigates compact spatiotemporal neural architectures [14] and video generalization across domains [15]. PRESENCIFY deliberately does not have presentation attack detection and claims no resilience against printouts or digital playback attacks. Biometric privacy studies prove that unencrypted facial templates can be reconstructed [16], [17]. Offline operation improves data containment but does not relieve regulatory compliance. Moreover, demographic fairness analysis shows that convenience samples are inadequate for validation of performance across diverse user populations [18].

III. METHODOLOGY

A. System architecture

Fig. 1 demonstrates the hardware-aware end-to-end processing pipeline starting from video acquisition and ending with SQLite database persistence. OpenCV captures webcam frames using native video backends of the operating system. MediaPipe BlazeFace locates facial bounding boxes and extracts six key landmarks in 3.5 ms using CPU. Affine transformation transforms the detected face into a 112 by 112 pixels canonical face in 0.8 ms. EdgeFace XS processes the face crop in 15.3 ms, producing normalized 512-dimensional embedding vector. SQLite persists student enrollment templates, session parameters, attendance log entries and audit trail of administrator actions. Modular Streamlit UI coordinates the processes of enrollment, live scanning, review queue management, and attendance exporting. Enrollment requires explicit user consent and UI persistently displays PAD_UNAVAILABLE status indicator. Table I summarizes the core components of the system together with their implementation caveats.

TABLE I
REPORTED IMPLEMENTATION COMPONENTS AND CAVEATS

Component	Implementation	Role / Caveat
Detector	MediaPipe BlazeFace, CPU	Locates faces in 3.5 ms; detection does not identify persons.
Recognizer	EdgeFace XS, $\gamma = 0.6$, 512-d	Generates 512-d embeddings; operates under CC BY-NC-SA 4.0 terms.
Decision policy	Accept ≥ 0.65; review ≥ 0.52	Below 0.52 indicates low confidence, not verified absence.
Temporal guard	Three consecutive frames	Filters transient camera flicker; not multi-person tracking.
Storage	SQLite	Persists enrollment templates, session logs and review events.
PAD	PAD_UNAVAILABLE	No defenses against physical printouts or screen replay attacks.

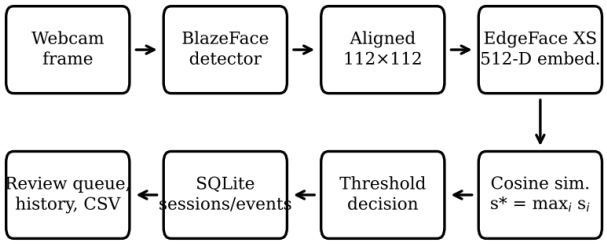


Fig. 1. PRESENCIFY implementation pipeline (CPU only). PAD is not part of the pipeline (PAD_UNAVAILABLE).

Attendance data trails the database operation of session-wise upserts of attendance records to ensure that detections of transient or sub-threshold presence do not overwrite any previously verified check-in timestamps. The administrative dashboard performs filtering of incidents pending verification by active lecture sessions and presents the instructor with candidate probe pictures to make the final judgment call. In classroom settings with webcams, varying lighting conditions, motion blur, and unenrolled attendees are common issues. When the recognizer outputs an UNKNOWN tag, this means that the confidence in matching did not meet operational thresholds, but not necessarily the absence of the student.

B. Scoring and decision policy

We define q as a probe embedding normalized to have unit norm and let g_i be the set of enrolled reference templates for active subjects. Pairwise cosine similarities and maximal match candidates are determined by Eq. 1. Cosine similarity provides an uncalibrated relative score rather than a true statistical probability of personal identity. Equation (2) defines the dual threshold strategy that is based on an upper acceptance threshold tau_a = 0.65 and a lower threshold for review tau_r = 0.52. Scores above the threshold tau_a lead to automatic enrollment, scores between the two thresholds [tau_r, tau_a) enter the instructor review queue, while scores below tau_r are marked as UNKNOWN. Runtime operation of the software entails three successive frames for confirmation of attendance; this feature

has not been taken into account in the benchmark test static evaluation.

$$s_i = q^T g_i, \quad s^* = \max_{\{i\}} s_i \quad (1)$$

$$\text{Decision}(q) = \{ \text{ACCEPT: } s^* \geq 0.65; \text{ REVIEW: } 0.52 \leq s^* < 0.65; \text{ UNKNOWN: } s^* < 0.52 \} \quad (2)$$

C. Evaluation protocol and data

The empirical audit utilizes 410 facial captures partitioned into four distinct experimental splits, as summarized in Table II. The gallery split reserves 40 captures representing 20 enrolled subjects (two reference images per individual). The validation split comprises 100 captures designated for threshold exploration, strictly separated from subsequent testing. The primary held-out test split contains 166 static probe images, structured as 100 genuine probes from enrolled students and 66 impostor probes from unenrolled individuals. An unused split of 104 captures is set aside to maintain complete separation between experimental phases.

TABLE II
CAPTURE PARTITIONS IN THE CONTROLLED EVALUATION (410 CAPTURES)

Partition	Captures	Evaluation Purpose
Gallery	40	Enrolled reference templates (2 captures per subject across 20 students)
Validation	100	Threshold exploration split; completely separated from test probes
Held out test	166	Primary evaluation split: 100 genuine probes + 66 impostor probes
Unused	104	Excluded captures; prevents leakage across experimental stages
Total	410	Sum fully accounted for across all experimental partitions

Genuine test probes originate from enrolled subjects captured during separate imaging sessions, ensuring a capture-disjoint evaluation. Impostor probes are sourced exclusively from unenrolled individuals, guaranteeing an open-set identity-disjoint test protocol. Method A establishes a baseline single-threshold policy, accepting probes with similarity >= 0.65 and rejecting all lower scores as UNKNOWN. Method B introduces the uncertainty band, passing ambiguous probes between 0.52 and 0.65 to Stage 2 review. In the evaluation script, Stage 2 assigns outcomes using ground-truth subject metadata, establishing an idealized upper bound rather than measuring real human performance.

A critical methodological caveat concerns the provenance of probe images for subject P001. The evaluation pipeline attempted live webcam acquisition. When capture yielded fewer than ten facial crops, synthetic photometric variants were generated from a single capture; when acquisition failed entirely, the system substituted the initial ten images of the Olivetti research benchmark. These benchmark fallback images overlap with the initial slice used for subject P002. Because saved run logs do not definitively record which capture path executed, evaluators cannot verify that all 410 images represent authentic, independent webcam captures.

The repository contains test configuration discrepancies between documentation versions. Earlier summaries report 38 unit tests, whereas updated test executions confirm 53 passing test cases. These documentation variations underscore the importance of auditing executable test scripts directly rather than relying on static documentation badges.

D. Metrics

System performance is quantified using three primary metrics defined in Equations (3), (4), and (5). The False Acceptance Rate (FAR) measures the proportion of unauthorized impostor probes accepted as enrolled attendees. The Identity False Rejection Rate (FRR_id) quantifies genuine student probes that are either outright rejected or misidentified as another enrolled peer. The Review Rate (R_review) calculates the proportion of total probe traffic directed into the teacher review queue. These formulations strictly distinguish binary attendance decisions from fine-grained identity verification.

$$FAR = (\text{impostor probes accepted as enrolled}) / N_{imp} \quad (3)$$

$$FRR_{id} = (N_{gen\ rejected} + N_{gen\ misidentified}) / N_{gen} \quad (4)$$

$$R_{review} = N_{review} / N_{total} \quad (5)$$

IV. RESULTS AND DISCUSSION

A. Stored results

Table III summarizes the stored experimental results across the 166 held-out test probes (100 genuine, 66 impostor). Fig. 2 visually illustrates genuine probe outcomes and persistent impostor leakage across evaluation stages.

TABLE III
STORED TEST RESULTS (166 PROBES: 100 GENUINE, 66 IMPOSTOR).
STAGE 2 IS ORACLE SIMULATED, NOT OBSERVED HUMAN PERFORMANCE.

Measure	A: direct	B: Stage 1	B: Stage 2†	Observed Impact
Correct identity accepts	65 / 100	65 / 100	77 / 100	+12 genuine rescued
Misidentifications	7 / 100	7 / 100	7 / 100	Remains isolated
Genuine rej. / queued	28 rejected	5 rej. + 23 q.	16 rejected	Cut from 35% to 16%
Impostor false accepts	18 / 66	18 / 66	18 / 66	Upper tail (≥ 0.65)
Impostor rejections	48 / 66	15 rej. + 33 q.	48 / 66	Maintained across stages
Review workload	0 / 166 (0%)	56 / 166 (33.7%)	56 audited	Quantified human burden
Identity Precision	0.7222	None	0.7549	+3.27% improvement
Identity Recall	0.6500	None	0.7700	+12.00% improvement
Identity F1 Score	0.6842	None	0.7624	+0.0782 gain
Binary F1 Score	0.7579	None	0.8317	+0.0738 gain
Overall Accuracy	68.07%	None	75.30%	+7.23% net gain

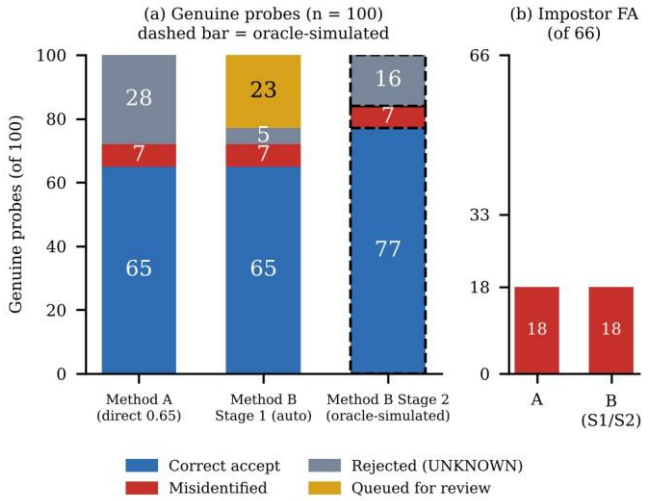


Fig. 2. Outcomes from the stored results. (a) Genuine probes (n = 100) per policy; the dashed outline Stage 2 bar is oracle simulated. (b) Impostor false accepts (n = 66) at s* ≥ 0.65, unchanged by review.

B. Automated stage

Under Method A, direct thresholding at 0.65 yields 65 correct genuine acceptances, 7 misidentifications, and 28 false rejections (FRR_id = 35.00%). Crucially, 18 of 66 impostor probes equal or exceed 0.65, resulting in an alarming 27.27% False Acceptance Rate. Because these high-scoring impostor probes match or exceed the automated acceptance cutoff, they bypass automated screening entirely. The review band cannot intercept them, demonstrating that static thresholding cannot eliminate confident proxy attendance attempts.

C. Review band and workload

The uncertainty review band [0.52, 0.65) intercepts 56 of the 166 test probes, generating a substantial 33.73% review burden. Queued items include 23 genuine probes and 33 borderline impostor probes (Fig. 3). Stage 1 automatically accepts 90 probes and rejects 20 probes without human intervention. While the review queue successfully intercepts 33 borderline impostor attempts, it requires instructors to manually inspect more than one-third of all attendance interactions.

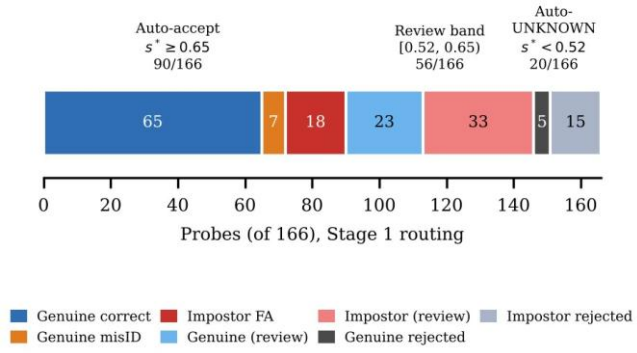


Fig. 3. Stage 1 routing of the 166 test probes under Method B, from stored counts: 90 probes are automatically accepted (65 correct, 7 misidentified genuine, 18 impostor false accepts), 56 are queued for review (23 genuine and 33 impostor) and 20 are automatically rejected (5 genuine, 15 impostor). Stage 2 is not shown because it is oracle simulated.

D. Simulated oracle review in Stage 2

Under oracle-simulated review, automated checking against ground-truth subject metadata recovers 12 genuine student probes and correctly rejects all 33 borderline impostor probes. Correct genuine verifications increase from 65 to 77, reducing false rejections to 16. This simulated recovery raises identity F1 from 0.6842 to 0.7624 and overall accuracy to 75.30%. Evaluators must recognize that these gains represent a theoretical mathematical upper bound rather than observed human instructor grading performance.

E. Consistency of the stored metrics

The stored summary metrics remain fully consistent with underlying probe frequency tallies. For Method A, 65 true positive accepts alongside 25 false positives produce an identity precision of 0.7222. Following simulated review, 77 true accepts alongside 25 false positives yield an identity precision of 0.7549. Binary attendance F1 reaches 0.7579 for Method A and climbs to 0.8317 under simulated review. Binary F1 scores should not be compared directly with identity F1 metrics because their operational definitions of positive cases differ.

F. Computational timing

Table IV details execution benchmarks gathered on an Intel Core i3-1115G4 processor without GPU acceleration. MediaPipe BlazeFace completes face detection in 3.5 ms, and affine alignment requires approximately 0.8 ms. EdgeFace XS extracts facial embeddings in 15.3 ms on average, yielding an aggregate computational latency of 19.6 ms per frame (approximately 51 FPS compute throughput). However, an earlier project document reports an average latency of 37 ms with peaks of 68 ms. Neither benchmark accounts for camera IO, Streamlit browser rendering, or multi-frame temporal buffering.

TABLE IV
REPORTED COMPUTATIONAL TIMINGS (INTEL CORE I3-1115G4, CPU ONLY).
COMPONENT TIMINGS, NOT END TO END WEBCAM THROUGHPUT.

Hardware Subsystem	Reported Value	Operational Caveat
EdgeFace XS pass	15.3 ms (mean)	Forward pass only; excludes frame capture
BlazeFace detector	3.5 ms (mean)	Detects face bounding box and 6 key landmarks
Affine alignment	0.8 ms (mean)	Warps cropped face to 112 × 112 canonical format
Aggregate execution	19.6 ms (≈51 FPS)	Total algorithm computation; excludes camera IO
Alternative report	37 ms avg., 68 ms max.	Unreconciled discrepancy found in older logs

G. Discussion

The dual decision thresholds partition separate regions of the similarity distribution. The upper cutoff tau_a distinguishes immediate passes from manual review candidates, while tau_r eliminates obvious non-matching faces. Because 18 impostors surpass 0.65, adjusting the review band cannot mitigate high-confidence proxy attendance leakage. The review queue functions as an operational triage buffer for borderline ambiguities rather than a universal security filter.

Borderline review queues contain more impostor probes (33) than genuine student captures (23). Consequently, instructors

allocate the majority of their review effort to rejecting unauthorized individuals rather than recovering legitimate students. Any empirical accuracy recovery for genuine students depends entirely on instructors making flawless visual verification judgments.

Existing attendance literature reports varied verification accuracies across bespoke hardware configurations and closed datasets [1]–[3]. Direct numerical score competition with those systems is methodologically invalid. PRESENCIFY instead contributes an open-set empirical audit documenting impostor leakage and human review workloads. Because standard metric matchers lack presentation attack detection [11], [13], high embedding similarity provides no guarantee against physical photograph spoofing.

H. Reproducibility and test status

Stored artifacts confirm that the four-stage evaluation script executed to completion with exit code 0. The automated test suite executes 53 passing test cases in 22.72 seconds following codebase refactoring. Earlier documentation records citing 38 tests reflect preliminary project stages. Git commit hashes b44205f and 0690d0a document verified codebase milestones. Future replications should execute the complete pipeline to verify individual probe scores directly.

I. What the evidence supports

Table V distinguishes claims directly supported by project artifacts from assumptions that lack empirical evidence. The prototype confirms feasible CPU-only inference but cannot guarantee flawless verification or physical spoof resilience.

TABLE V
SCOPE OF THE EVIDENCE

Supported Cautiously	Not Supported by Evidence
Working prototype workflow for registration, webcam scanning, attendance events and history export.	Production deployment across enterprise campus networks or unmonitored gates.
Dual threshold recognition with human review executed smoothly on commodity CPU hardware.	Guaranteed recognition accuracy or complete elimination of proxy attendance.
Manual review of borderline cases can rescue genuine false absences under ideal conditions.	Empirical validation of real world teacher review speed or human accuracy.
Transparent user interface disclosure of PAD_UNAVAILABLE status across all screens.	Presentation attack resistance, 3D mask spoof defense or photo replay detection.

J. Limitations

Multiple methodological constraints qualify the empirical scope of this study. Foremost among these is the complete absence of presentation attack detection (PAD_UNAVAILABLE), leaving the system vulnerable to printed photographs or screen replay attacks. Furthermore, 18 of 66 impostors achieved high-confidence automated acceptance, indicating that static thresholds cannot prevent determined proxy attendance. The Stage 2 review evaluation reflects an oracle simulation using ground-truth metadata rather than empirical classroom trials with practicing teachers. Additionally, the small 41-subject cohort lacks demographic diversity, challenging illumination, and wide pose variations. The capture provenance for subject P001 remains unresolved across stored logs. Finally, the three-frame temporal guard was omitted from static probe testing, reported latency benchmarks exhibit minor documentation variance, and facial biometric templates demand stringent legal safeguards, affirmative user consent, and transparent data deletion procedures [16], [17].

K. Threats to validity

Internal validity threats involve omitting temporal consistency checks during static testing and relying on ground-truth metadata during simulated reviews. If subject P001 utilized benchmark fallback images, genuine and impostor partitions share overlapping image slices. External validity is constrained because a small static evaluation cohort cannot capture the operational turbulence of crowded campus lecture halls. Reporting discrepancies in benchmark latencies and test counts require ongoing documentation synchronization. Moreover, the evaluation omitted formal confidence intervals and hypothesis testing.

Construct validity faces limitations because the investigation audits only a single pair of fixed decision thresholds. Raw cosine similarity distributions shift across different camera hardware, optical lens distortion, and ambient room lighting. Thresholds calibrated on one specific workstation cannot be assumed to function reliably in another environment. In the absence of human observer trials, simulated reviews cannot model real instructor fatigue, visual bias, or verification mistakes.

Threshold transportability presents an ongoing operational hurdle for edge biometric systems deployed across educational campuses. A similarity cutoff that separates identities under controlled lighting will drift when room illumination shifts. Until systems are evaluated across diverse physical classrooms, static benchmark numbers remain provisional engineering observations.

L. Privacy and responsible use

Facial embeddings constitute sensitive biometric data subject to rigorous legal and ethical governance [16], [17]. PRESENCIFY mandates affirmative student consent during registration and never persists raw video frames to disk. Production deployments must implement pseudonymous identifiers, database encryption, and automated record deletion policies. The EdgeFace XS model checkpoint carries a noncommercial research license (CC BY-NC-SA 4.0). Academic institutions must respect these licensing terms before sharing code or deploying systems commercially.

V. CONCLUSION

PRESENCIFY demonstrates that functional facial attendance workflows can execute locally on commodity laptop processors without dedicated graphics cards. Combining BlazeFace and EdgeFace XS achieves a 19.6 ms computational budget on CPU hardware. However, static decision thresholds permit 27.27% of unauthorized impostors to pass automated verification. Introducing an uncertainty review band intercepts borderline impostors but imposes a manual inspection workload on 33.73% of all attendance trials.

Simulated oracle review recovers genuine students only under the unrealistic assumption that instructors make zero misidentifications. The prototype transparently discloses its PAD_UNAVAILABLE status across all interfaces to prevent unjustified reliance on spoof resistance. Future research must evaluate active presentation attack detection modules, assemble

diverse classroom datasets, and conduct formal human-in-the-loop teacher verification trials.

As a college mini-project, PRESENCIFY succeeds by delivering a reproducible, inspectable edge biometric prototype. Rather than making unsubstantiated claims of high accuracy or security, the study transparently documents real edge bottlenecks, algorithmic trade-offs, and essential operational caveats.

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